Polynomial Auditing For Black-Box Differentially Private Mechanism

Kaixin Zheng
David R. Cheriton School of Computer Science
University of Waterloo
Waterloo, ON N2L 5Z5
k59zheng@uwaterloo.ca

Abstract

We study black-box auditing of differential privacy via global approximation of the trade-off function. Unlike prior pointwise auditing methods, our approach fits a conservative upper envelope and converts it into an audited (ε, δ) guarantee through tangent geometry. Concretely, after rotating the trade-off plane, we fit an even convex polynomial subject to structural constraints motivated by valid trade-off functions. We evaluate the method on the Gaussian mechanism and a toy DP-SGD setting with analytic ground truth. The results show that the method gives reasonably tight estimates in clean settings and remains informative under noisy observations. We also identify the main limitations of the approach, including tail sensitivity, noise in estimated trade-off pairs, and structural mismatch for asymmetric mechanisms.

1 Introduction

Auditing privacy guarantees has become increasingly important, especially as machine learning models are now deployed in sensitive domains such as healthcare, finance, and government services [2, 7, 12]. This problem is particularly challenging in the black-box setting, where the auditor has access only to model outputs. A large portion of prior work either requires access to model parameters [10], internal states [8], or knowledge of the parametric family of the underlying mechanism [11].

For black-box auditing, a substantial body of prior work studies privacy through pointwise, hypothesis-testing-based estimates [3, 11]. In this framework, one compares the observed true positive rate with the theoretically guaranteed true positive rate at a given false positive rate, or equivalently studies the trade-off between type-I and type-II errors at selected operating points. However, because these approaches are inherently pointwise, it is generally difficult to convert them into standard (ε, δ)-DP guarantees [6] in a direct and tight way.

Motivated by the f-DP framework in [4], several recent works have explored the use of f-DP for privacy auditing. These include approaches based on canary construction and comparison between empirical output distributions and theoretically guaranteed distributions [13], as well as membership-inference-based auditing methods [9, 14]. Nevertheless, existing methods still largely rely on pointwise comparisons or mechanism-specific structures, which makes it difficult to obtain a global and conservative characterization of privacy directly from black-box observations.

In this work, we propose a conservative auditing method based on fitting an upper envelope of the trade-off function. Our key idea is to approximate a trade-off curve that lies above the true mechanism on the trade-off plane, and then convert this fitted curve into an audited (ε, δ) guarantee. Under suitable regularity assumptions, including smoothness and symmetry of the trade-off function, this procedure yields a meaningful and often tight privacy audit.

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Our main contributions.

- We propose a conservative curve-fitting framework for black-box privacy auditing. Instead of estimating privacy only at isolated operating points, our method fits a global trade-off function with structural constraints motivated by the theory of trade-off functions.
- We introduce a polynomial upper-envelope parameterization in rotated coordinates, which naturally incorporates symmetry and enables convexity. This provides a practical way to construct a conservative approximation to the true trade-off curve from noisy empirical samples.
- We show how to convert the fitted trade-off curve into audited (ε, δ) guarantees through a tangent-based characterization. This gives a direct bridge from empirical black-box observations to standard differential privacy guarantees.
- We empirically evaluate the proposed method on Gaussian mechanism and DP-SGD to show that it can recover reasonably tight privacy estimates in clean settings and remains informative in noisy black-box settings.
- We further analyze the main limitations of the approach, including sensitivity to tail accuracy, noise in empirical trade-off samples, and the difficulty of auditing realistic mechanisms such as DP-SGD in the small-δ regime.

2 Preliminaries

2.1 (ϵ, δ)-DP

Definition 2.1. Differential privacy ([5]) A mechanism M is said to be (ϵ, δ)-differentially private if, for every measurable set T and every pair of neighboring datasets S, S', it holds that

$$\Pr[M(S) \in T] \leq e^{\epsilon} \Pr[M(S') \in T] + \delta.$$

2.2 f-DP

We briefly review the hypothesis-testing view of privacy and the definitions of trade-off functions and f-DP following the standard presentation in [3] along with original definition and theorems from [4].

Hypothesis testing. Let P and Q be two probability distributions on the same sample space, and suppose we wish to distinguish between the two hypotheses

$$H_0 : X \sim P, \qquad H_1 : X \sim Q.$$

A (possibly randomized) hypothesis test is a measurable function

$$g : \mathcal{X} \to \{0, 1\},$$

where $g(X) = 1$ means rejecting H_0. The corresponding type-I and type-II errors are defined by

$$\alpha(g) := \Pr_{X \sim P}[g(X) = 1], \qquad \beta(g) := \Pr_{X \sim Q}[g(X) = 0].$$

For a prescribed type-I error level $\alpha \in [0, 1]$, one is interested in the smallest achievable type-II error among all tests satisfying $\alpha(g) \leq \alpha$.

Definition 2.2. Trade-off function The trade-off function associated with P and Q is defined by

$$T(\alpha) := \inf\big\{\beta(g) : g \text{ is a test with } \alpha(g) \leq \alpha\big\}, \qquad \alpha \in [0, 1].$$

Thus, $T(\alpha)$ represents the smallest possible type-II error when the type-I error is constrained to be at most α.

Definition 2.3. f-differential privacy Let M be a randomized mechanism, and let S, S' be neighboring datasets. Denote by P and Q the output distributions of $M(S)$ and $M(S')$, respectively. The corresponding trade-off function is denoted by T.

We say that M is *f-differential private* if for every pair of neighboring datasets S, S',

$$T(\alpha) \geq f(\alpha), \qquad \forall \alpha \in [0, 1].$$

Proposition 2.1. The following statements hold.

1. A function $f : [0,1] \to [0,1]$ is a trade-off function if and only if f is convex, continuous, non-increasing, and satisfies
$$f(x) \le 1 - x, \qquad \forall x \in [0,1].$$
2. If a mechanism M is f-DP, then M is also f_S-DP, where
$$f_S = \max\{f, f^{-1}\},$$
and the inverse function f^{-1} is defined by
$$f^{-1}(\alpha) := \inf\{t \in [0,1] : f(t) \le \alpha\}, \qquad \forall \alpha \in [0,1].$$

Corollary 2.1. Trade-off function of (ϵ, δ)-DP. The mechanism M is (ϵ, δ)-DP mechanism if and only if its associated trade-off function is

$$f_{\epsilon,\delta}(\alpha) = \max\Big\{0,\ 1 - \delta - e^{\epsilon}\alpha,\ e^{-\epsilon}(1 - \delta - \alpha)\Big\}.$$

Corollary 2.2. Gaussian trade-off function. For $\mu \ge 0$, define

$$G_\mu(\alpha) = \Phi\big(\Phi^{-1}(1-\alpha) - \mu\big)\,.$$

Theorem 2.1. Central limit theorem of f-DP The operator $\otimes$ denotes the composition of trade-off functions. Repeated composition of (ϵ_i, δ_i)-DP trade-off functions converges to a Gaussian trade-off function composed with a $(0, 1 - e^{-\delta})$-DP mechanism:

Assume

$$\sum_{i=1}^{n} \epsilon_{ni}^2 \to \mu^2, \qquad \max_{1 \le i \le n} \epsilon_{ni} \to 0,$$

and

$$\sum_{i=1}^{n} \delta_{ni} \to \delta, \qquad \max_{1 \le i \le n} \delta_{ni} \to 0$$

for some nonnegative constants μ, δ as $n \to \infty$. Then

$$f_{\epsilon_{n1},\delta_{n1}} \otimes \cdots \otimes f_{\epsilon_{nn},\delta_{nn}} \to G_\mu \otimes f_{0,1-e^{-\delta}}$$

uniformly over $[0,1]$ as $n \to \infty$.

3 Method

Theorem 2.1. motivates our approach: if the lower-bound privacy profile under composition converges to a Gaussian limit, then a similar Gaussian-type phenomenon may also be expected on the auditing side when constructing conservative upper bounds. This suggests approximating the true mechanism by a structured function class satisfying the properties in Proposition 2.1., and then using the resulting upper envelope for privacy auditing. Even without composition, the method holds if the single mechanism has appropriate symmetry in hypothesis testing.

Auditing objective. Given a fixed $\delta \in (0,1)$, let $\varepsilon_{\mathrm{true}}(\delta)$ denote the smallest value such that a mechanism M is $(\varepsilon_{\mathrm{true}}(\delta), \delta)$-DP with respect to neighboring datasets S and S'. Our goal is to infer a value $\varepsilon^\star$ satisfying

$$\varepsilon^\star \le \varepsilon_{\mathrm{true}}(\delta),$$

so that $\varepsilon^\star$ serves as a lower bound on the true privacy parameter at the target level δ.

Auditor access model. The auditor does not know the internal form of M. Instead, it only observes a finite collection of sampled or estimated trade-off points

$$\{(\alpha_i, \beta_i)\}_{i=1}^{N}$$

associated with the output distributions $M(S)$ and $M(S')$. In our experiments, these points are either noiseless evaluations from an analytic trade-off curve or noisy estimates obtained from KDE.

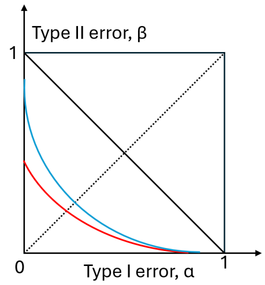

Figure 1: True mechanism (●) and fitted curve (●) on trade-off plane.

Structured surrogate fitting. We fit a symmetric polynomial surrogate to the trade-off curve after rotating the (α, β) coordinates by $\pi/4$. Let (x_i, y_i) denote the rotated coordinates of (α_i, β_i). We parameterize the fitted curve as

$$g(x) = \sum_{j=0}^{M} c_{2j} x^{2j},$$

where only even powers are used to enforce symmetry.

The coefficients are obtained by solving a constrained optimization problem. The fit is required to satisfy the basic geometric properties in the rotated coordinate system: (i) convexity; (ii) the lower and upper envelope constraints induced by the indistinguishable line and the unit box; (iii) the fitted curve is encouraged to lie above the sampled data pairs by imposing a soft penalty on points that exceed the fitted curve. To reduce the influence of regions that are less informative for privacy extraction, we use a weighted objective that places more emphasis on tail behavior and solve

$$\min_{c_0,\dots,c_{2M}} \sum_{i=1}^{N} w(x_i)\Big(r_i^2 + \lambda\, \mathrm{ReLU}(r_i)^2\Big),$$

where $r_i = y_i - g(x_i)$, subject to the structural constraints above. $\lambda \geq 0$ is a hyperparameter. The full constrained formulation and analysis are given in Appendix A.

Extracting $\varepsilon(\delta)$. After fitting g, we recover a privacy estimate from the tangent geometry of the fitted curve. For each admissible tangent point $x_0 \leq 0$, the fitted curve induces a pair $(\delta(x_0), \varepsilon(x_0))$. Given a target δ, we solve for the corresponding tangent point and report the resulting value as the audited estimate

$$\hat{\varepsilon}(\delta).$$

The explicit formulas for $\delta(x_0)$ and $\varepsilon(x_0)$ are deferred to Appendix A.

Visual illustration. As shown in Fig. 1, our approach aims to find the tightest convex envelope above the sampled data points.

4 Experiments

4.1 Gaussian mechanism

We use the Gaussian mechanism as a proof-of-concept example. In this experiment, the sensitivity is $\Delta = 2$ and the noise standard deviation is $\sigma = 1$, so the trade-off function is available in closed form:

$$G_\mu(\alpha) = \Phi\big(\Phi^{-1}(1-\alpha) - \mu\big)\,, \qquad \mu = \Delta/\sigma = 2. \tag{1}$$

This allows us to directly compare the fitted auditing result against the ground-truth (ϵ, δ) relationship.

Effect of polynomial degree. We first sample α uniformly over $[10^{-12}, 1-10^{-12}]$ and study the effect of the polynomial degree M. Table 1 shows that increasing M removes optimistic violations ($M = 5$, $\delta = 10^{-3}$), but also makes the fitted ϵ more conservative, especially in the small-δ regime. For example, when $M = 5$ and $\delta = 10^{-3}$, the fitted ϵ slightly exceeds the true value, whereas this violation disappears for $M = 10$ and $M = 20$. However, the corresponding estimates for smaller δ become noticeably looser. This suggests that increasing polynomial degree mainly improves feasibility of the constrained fit, rather than tail accuracy itself.

M	N	δ	True ϵ	Fitted ϵ
5	1000	10^{-2}	5.9979	5.9743
5	1000	10^{-3}	7.5813	7.6894
5	1000	10^{-4}	8.8769	8.5443
5	1000	10^{-5}	9.9973	8.8403
10	1000	10^{-2}	5.9979	5.7708
10	1000	10^{-3}	7.5813	6.2947
10	1000	10^{-4}	8.8769	6.4040
10	1000	10^{-5}	9.9973	6.4275
20	1000	10^{-2}	5.9979	5.8081
20	1000	10^{-3}	7.5813	6.3624
20	1000	10^{-4}	8.8769	6.4781
20	1000	10^{-5}	9.9973	6.5030

Table 1: Comparison between true and fitted ϵ values for different polynomial degrees M, using $N = 1000$ uniformly sampled points.

Effect of sample size. To isolate the role of sample size, we fix a highly expressive polynomial class with $M = 50$ and vary the number of sampled points N. Table 2 shows that increasing N from 10 to 600 removes the remaining optimistic behavior, while a further increase from 600 to 1000 produces almost no change. Therefore, once the constrained fit becomes stable, simply adding more uniformly sampled points does not substantially improve the final audited values. This indicates that the main bottleneck is not only the amount of fitting data, but also how the sampled points are distributed along the trade-off curve.

M	N	δ	True ϵ	Fitted ϵ
50	10	10^{-2}	5.9979	6.5298
50	10	10^{-3}	7.5813	7.4031
50	10	10^{-4}	8.8769	7.5755
50	10	10^{-5}	9.9973	7.6118
50	600	10^{-2}	5.9979	5.8049
50	600	10^{-3}	7.5813	6.3566
50	600	10^{-4}	8.8769	6.4718
50	600	10^{-5}	9.9973	6.4965
50	1000	10^{-2}	5.9979	5.8078
50	1000	10^{-3}	7.5813	6.3618
50	1000	10^{-4}	8.8769	6.4774
50	1000	10^{-5}	9.9973	6.5022

Table 2: Comparison between true and fitted ϵ values for $M = 50$ under different numbers of sampled points N.

Effect of sampling strategy. To better resolve the tails, we replace uniform sampling in α with a deterministic tail-heavy grid induced by the quantiles of a symmetric Beta distribution. Specifically, let

$$u_i \in [\alpha_{\min}, 1-\alpha_{\min}], \qquad \alpha_{\min} = 10^{-12},$$

be uniformly spaced, and define

$$\alpha_i = \alpha_{\min} + (1-2\alpha_{\min})\, Q_{\mathrm{Beta}(a,a)}(u_i),$$

where $Q_{\mathrm{Beta}(a,a)}$ denotes the quantile function of the $\mathrm{Beta}(a, a)$ distribution with $a = 0.35$. Tables 3 and 4 show that the effect of tail-heavy sampling depends strongly on model capacity. For $M = 50$, tail-heavy sampling does not significantly improve the already conservative fit. In contrast, for $M = 5$, it removes the optimistic violation at $\delta = 10^{-3}$, although this comes at the cost of more conservative estimates for smaller δ. Overall, these results suggest that non-uniform sampling is most useful in under-parameterized regimes, where it reduces violation risk.

Sampling	M	Num. Points	δ	True ϵ	Fitted ϵ
Uniform	50	1000	10^{-2}	5.9979	5.8078
Uniform	50	1000	10^{-3}	7.5813	6.3620
Uniform	50	1000	10^{-4}	8.8769	6.4776
Uniform	50	1000	10^{-5}	9.9973	6.5024
Tail-heavy	50	1000	10^{-2}	5.9979	5.8384
Tail-heavy	50	1000	10^{-3}	7.5813	6.4483
Tail-heavy	50	1000	10^{-4}	8.8769	6.5890
Tail-heavy	50	1000	10^{-5}	9.9973	6.6246

Table 3: Comparison between uniform and tail-heavy sampling for $M = 50$ and $N = 1000$.

Sampling	M	Num. Points	δ	True ϵ	Fitted ϵ
Uniform	5	1000	10^{-2}	5.9979	5.9741
Uniform	5	1000	10^{-3}	7.5813	7.6857
Uniform	5	1000	10^{-4}	8.8769	8.5370
Uniform	5	1000	10^{-5}	9.9973	8.8315
Tail-heavy	5	1000	10^{-2}	5.9979	5.9673
Tail-heavy	5	1000	10^{-3}	7.5813	7.5586
Tail-heavy	5	1000	10^{-4}	8.8769	8.2932
Tail-heavy	5	1000	10^{-5}	9.9973	8.5417

Table 4: Comparison between uniform and tail-heavy sampling for $M = 5$ and $N = 1000$.

Key findings. Overall, these ablations suggest that the main challenge is not merely to fit the trade-off curve globally, but to allocate enough approximation and constraint resolution to the tail region, which ultimately determines the quality of the audited (ϵ, δ) guarantee at small δ.

4.2 Auditing DP-SGD with noiseless data

We next evaluate our method on differentially private stochastic gradient descent (DP-SGD), using the experimental setting of [3] as our reference benchmark. In particular, we follow their one-dimensional quadratic-loss setup, where the neighboring datasets are chosen as

$$D = (0, \ldots, 0), \qquad D' = (1, 0, \ldots, 0),$$

with database size $r = 10$. The loss function is

$$\ell(\theta, x_i) = \frac{1}{2}(\theta - x_i)^2,$$

with initial model $\theta_0 = 0$ and parameter space $\Theta = \mathbb{R}$. The remaining DP-SGD parameters are fixed to $\sigma = 0.2$, $\rho = 0.2$, $\tau = 10$, and $m = 5$. The corresponding trade-off curve can be derived analytically as in Appendix B.

In this experiment, the fitting data are generated by sampling α uniformly over $[10^{-12}, 1 - 10^{-12}]$. For the target privacy level $\delta = 10^{-5}$, the theoretical worst-case value is $\epsilon = 8.6665$. We use $N = 1000$ sampled points and an even polynomial of degree $M = 50$.

Conservative auditing due to structural limitation. As shown in Figure 2, our auditing method returns $\epsilon = 3.1909$, which is valid but highly conservative compared to the true worst-case value $\epsilon = 8.6665$.

Figure 2 shows that the true DP-SGD trade-off curve is not symmetric in the trade-off plane. Our method, however, searches for a symmetric convex polynomial that remains above the true curve. For this particular non-symmetric mechanism, this creates a structural limitation: the fitted upper envelope is effectively constrained by the right tail. As a result, the recovered auditing value is governed by the right-side geometry rather than by the true worst-case side of the mechanism.

The right-side ϵ of the true mechanism is 3.6739, while our fitted polynomial gives $\epsilon = 3.1909$. Thus, relative to the right-side ϵ, the recovered value achieves about 87% accuracy. In contrast, the true worst-case value, corresponding to the left tail, is $\epsilon = 8.6665$. Therefore, the discrepancy is not primarily caused by numerical error in the fitting procedure, but by the mismatch between the asymmetric shape of the true trade-off curve and the symmetric convex function class imposed by our method.

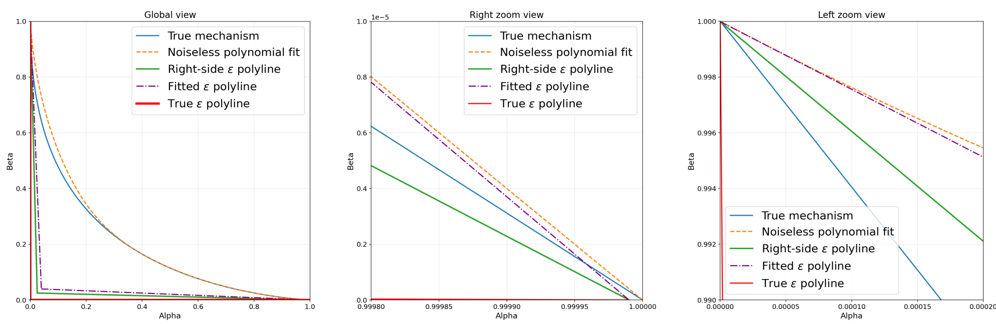

Figure 2: Comparison of true mechanism DP-SGD, fitted polynomial curve, true ϵ polyline, fitted ϵ polyline, and right-side ϵ polyline. Global, right-tail zoom, and left-tail zoom views with different α range. Corresponding ϵ value: true ϵ polyline, 8.665; right-side ϵ polyline, 3.6739; fitted ϵ polyline, 3.1909.

4.3 Auditing DP-SGD with noisy data

In this section, we audit the same DP-SGD mechanism as in Section 4.2, but now the (α, β) pairs are generated from kernel density estimation (KDE) following the procedure in [3], using $n_1 = 10^6$ samples.

Polynomial fitting remains reasonably robust under noisy data. As shown in Table 5, we report the auditing results for the first 20 iterations, that is, the first 20 independently generated KDE datasets. The resulting accuracy is comparable to that obtained in the noiseless setting. Moreover, this performance (mean relative accuracy 36%) is still substantially better than the black-box DP-SGD baseline reported in [11], whose relative accuracy is around 20% at $\delta = 10^{-5}$. At the same time, we emphasize that our experimental setup does not exactly match that of [11], so this comparison should be interpreted with caution.

Quantity	Value
Target δ	10^{-5}
Theoretical ϵ	8.6665
Mean fitted ϵ	3.1133
Standard deviation of fitted ϵ	0.1199
Mean relative accuracy	0.3592
Best iteration	iter 3
Best fitted ϵ	3.3295
Worst iteration	iter 8
Worst fitted ϵ	2.9110

Table 5: Summary of fitted ϵ values across 20 KDE iterations for DP-SGD at $\delta = 10^{-5}$.

Weighted optimization significantly improves utility Under the same target value of δ, we additionally remove the weights from the optimization objective, and summarize the results in Table 6.

We observe that the recovered ϵ becomes even more underestimated, by roughly 10% on average relative to the weighted version. This suggests that weighting plays an important role in preserving utility in the auditing task, especially in the tail-sensitive regime. More broadly, these results indicate that future work should focus both on improving data quality near the tails and on incorporating more suitable structural priors to better capture tail geometry.

Key findings. Our proposed method is reasonably robust to noise in the input data. Under the noisy setting, the main bottlenecks remain the asymmetric shape of the true mechanism, the quality of the tail data, and the difficulty of fitting the tail region accurately. This is consistent with the goal of our method: we aim primarily to recover a reliable (ϵ, δ) guarantee, rather than to detect violations over the entire trade-off curve.

Quantity	Value
Target δ	10^{-5}
Theoretical ϵ	8.6665
Mean fitted ϵ	2.8572
Standard deviation of fitted ϵ	0.1671
Mean relative accuracy	0.3297
Best iteration	iter 1
Best fitted ϵ	3.2615
Worst iteration	iter 2
Worst fitted ϵ	2.7141

Table 6: Summary of fitted ϵ values across 20 KDE iterations for DP-SGD at $\delta = 10^{-5}$ after removing the weights from the objective function.

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A Method details

A.1 Optimization problem formalization

As illustrated in Fig. 3, for each point (α, β) in the original coordinate system, we introduce the rotated coordinates

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix}, \qquad \theta = \frac{\pi}{4}. \tag{2}$$

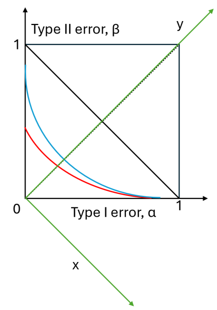

Figure 3: True mechanism (●) and fitted curve (●) on trade-off plane and rotated coordinate (●)

Thus, the observed trade-off pairs $\{(\alpha_i, \beta_i)\}_{i=1}^N$ are converted into rotated samples $\{(x_i, y_i)\}_{i=1}^N$. In the rotated coordinate system, we represent the trade-off curve by an even polynomial

$$y = g(x) := \sum_{j=0}^{M} C_{2j} x^{2j}. \tag{3}$$

Only even powers are used in order to enforce symmetry. Here, $2M$ is the maximum polynomial degree.

We require the fitted curve to preserve the basic geometric structure of a valid trade-off curve in the rotated coordinate system. First, convexity is imposed through

$$g''(x) \geq 0, \qquad \forall |x| \leq \frac{1}{\sqrt{2}}, \tag{4}$$

which is equivalent to

$$\sum_{j=1}^{M} 2j(2j-1) C_{2j} x^{2j-2} \geq 0, \qquad \forall |x| \leq \frac{1}{\sqrt{2}}. \tag{5}$$

Second, the fitted curve must lie within the feasible geometric region:

$$g(x) \leq \frac{1}{\sqrt{2}}, \qquad \forall |x| \leq \frac{1}{\sqrt{2}}, \tag{6}$$

$$g(x) \geq x, \qquad \forall |x| \leq \frac{1}{\sqrt{2}}. \tag{7}$$

To rewrite the problem in a simpler form, let

$$t = x^2, \qquad t_i = x_i^2, \qquad r_i = y_i - g(x_i).$$

Let $w : \mathbb{R} \to \mathbb{R}$ be a predefined monotone increasing weight function. In our default setting, we use

$$w(t) = 1 + 50 \left(\frac{t}{0.5} \right)^2.$$

The fitting problem is then formulated as

$$\begin{aligned}
\min_{\{C_0, \dots, C_{2M}\}} \quad & \sum_{i=1}^{N} w(t_i) \Big(r_i^2 + \lambda \, \mathrm{ReLU}(r_i)^2 \Big) \\
\text{s.t.} \quad & \sum_{j=1}^{M} 2j(2j-1) C_{2j} t^{j-1} \geq 0, \qquad \forall t \in [0, 1/2], \\
& \sqrt{t} \leq \sum_{j=0}^{M} C_{2j} t^j \leq \frac{1}{\sqrt{2}}, \qquad \forall t \in [0, 1/2].
\end{aligned}$$

Here, the soft penalty term $\mathrm{ReLU}(r_i)^2$ is introduced to discourage fitted curves that fall below the observed samples, while avoiding excessive sensitivity to outlying points that may otherwise distort the overall fitted shape.

Once the fitted trade-off curve $y = g(x)$ is obtained, the corresponding (δ, ϵ) values can be recovered from tangent geometry. For each tangent point $x_0 \le 0$, we define

$$\delta(x_0) = 1 - \sqrt{2}\,\frac{g(x_0) - g'(x_0)x_0}{1 + g'(x_0)}, \qquad \epsilon(x_0) = \ln\frac{1 - g'(x_0)}{1 + g'(x_0)}. \tag{8}$$

Therefore, for a prescribed target value of δ, we may solve for the corresponding tangent point x_0 and recover the associated auditing estimate ϵ.

A.2 Convergence analysis

We do not provide a formal analysis here. If the true mechanism is smooth and the sampling in the (x, y) coordinate system is sufficiently regular, then the problem is related to polynomial approximation of a convex function under shape constraints. For functions with limited smoothness, one may heuristically expect an L_∞ error on the order of $O(1/M)$ in the polynomial degree M, although establishing such a bound for our constrained optimization problem is beyond the scope of this project. This heuristic is consistent with classical approximation theory; see, e.g., [1] for Jackson-type results showing that polynomial approximation rates depend on smoothness.

B DP-SGD experiment details

As stated in [3], our DP-SGD experiment is

$$\theta^* = \arg\min_{\theta\in\Theta} L_x(\theta), \qquad L_x(\theta) = \frac{1}{r}\sum_{i=1}^{r} \ell(\theta, x_i).$$

Algorithm 1 DP-SGD for the Toy Problem

1: **Input:** Dataset $x = (x_1, \ldots, x_r)$, initial point $\theta_0 = 0$
2: **Input:** Learning rate ρ, batch size m, horizon τ, noise scale σ
3: **Output:** Final iterate θ_τ
4: **for** $t = 0, \ldots, \tau - 1$ **do**
5:     Sample a mini-batch $I_t \subseteq \{1, \ldots, r\}$ uniformly at random with $|I_t| = m$
6:     Compute

$$\frac{1}{m}\sum_{i\in I_t}(\theta_t - x_i)$$

7:     Sample noise $Z_t \sim \mathcal{N}(0, \sigma^2)$
8:     Update

$$\theta_{t+1} \leftarrow \theta_t - \rho\left(\frac{1}{m}\sum_{i\in I_t}(\theta_t - x_i) + Z_t\right)$$

9: **end for**
10: **return** θ_τ

This set-up comes with closed-form true mechanism.

$$T_{\mathrm{SGD}}(\alpha) = \sum_{I\subset\{1,\ldots,\tau\}} \frac{1}{2^\tau}\,\Phi\Big(\Phi^{-1}(1-\alpha) - \frac{\mu_I}{\bar\sigma}\Big),$$
$$\mu_I := \sum_{t\in I}\frac{\rho(1-\rho)^{\tau-t}}{m}, \qquad \bar\sigma^{\,2} := \frac{\rho^2\sigma^2\big(1-(1-\rho)^{2\tau}\big)}{1-(1-\rho)^2}.$$

where

$$\sigma = 0.2,\ \rho = 0.2,\ \tau = 10,\ m = 5.$$